

CASE STUDY CONTENT ARCHITECTURE

Flood Mapping Valencia

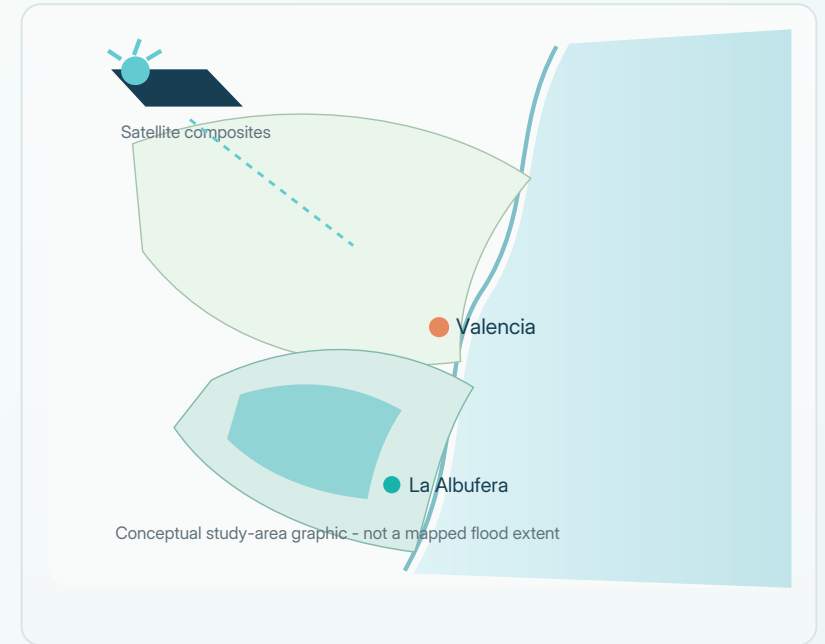
A web-ready narrative for explaining the October 29, 2024 Valencia flood through multi-temporal Sentinel-2 imagery, NDWI water mapping, and the broader spectral logic of NIR-SWIR disturbance indices.

01 Overview

02 Scope & Data

03 Analysis

04 Potential Further Study



5 time windows

Two pre-event, one immediate post-event, two recovery periods.

NDWI first

Green vs NIR contrast is the primary water-mapping signal in the source analysis.

NBR logic, carefully used

NIR-SWIR response explains disturbance sensitivity; wildfire severity classes are not transferred to flood mapping.

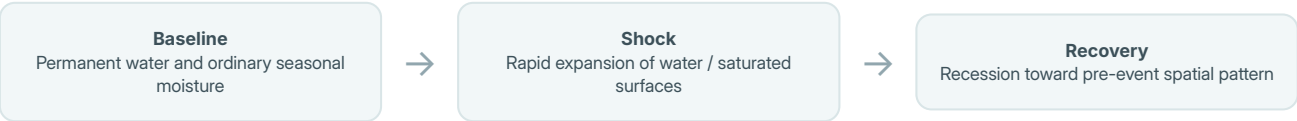
From an extreme rainfall event to a spatial recovery story

The case study asks a simple visual question: **how did the surface-water pattern around Valencia change before the flood, immediately after it, and through 2025?**

The October 29, 2024 event was driven by an intense DANA-related rainfall episode. AEMET reported 771.8 mm in the civil day at Turis, including about 185 mm in one hour. Copernicus Emergency Management Service (CEMS) was activated to map flood extent and impacts across the Valencia region.

CASE-STUDY FRAMING

Remote sensing is useful here because the footprint is regional and the response evolves quickly. Repeated satellite observations allow the narrative to shift from a one-day disaster snapshot to a pre-event baseline, event expansion, and recovery sequence.



29 Oct

Date of the most severe Valencia flood impacts in the 2024 episode.

771.8 mm

Civil-day rainfall reported at Turis by AEMET.

>53,000 ha

Area CEMS reported affected in Valencia province by Oct 31, 2024.

Sentinel-2

Multispectral optical imagery used for the source flood-mapping workflow.

WEBSITE-READY BLOCK

Suggested overview interaction

A short event timeline with a single hero statistic, followed by a compact animation showing "baseline - flood expansion - recovery." Keep the first screen interpretive rather than technical.

Define the geography, sensor, time windows, and indices before showing results

STUDY AREA

Valencia and the low-lying coastal plain to the south

The source interpretation emphasizes La Albufera, surrounding rice-growing agricultural lowlands, and channels draining toward the coast. These are locations where water and saturated soils create a strong spectral signal after the flood.

DATA SOURCE

Sentinel-2 Level-2A surface reflectance

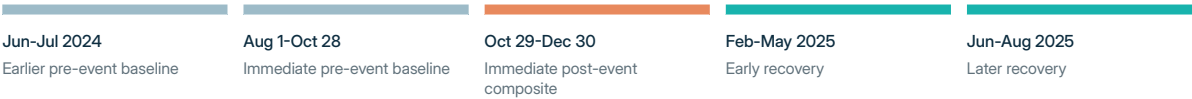
The workflow uses the harmonized Sentinel-2 surface-reflectance collection in Google Earth Engine. Sentinel-2's MSI has 13 bands; the relevant ones here span visible green, near-infrared, and short-wave infrared.

Band	Role	Center	Native res.
B3	Green	560 nm	10 m
B8	NIR	842 nm	10 m
B11	SWIR1	1610 nm	20 m
B12	SWIR2	2190 nm	20 m

TEMPORAL DESIGN

Five analysis windows

The uploaded narrative calls these “four temporal windows” but enumerates five. This case-study structure treats them as five distinct windows.



PROCESSING LOGIC



The source figures show mean composites in Earth Engine. For a production analysis, cloud and shadow handling should be explicitly documented because residual cloud can strongly bias water indices.

WEBSITE-READY BLOCK

Suggested data interaction

A compact “data at a glance” row for bands and resolution, plus a five-stop time selector. Keep processing details expandable so the main story remains readable.

NDWI maps water; NIR-SWIR indices explain a broader disturbance signal

PRIMARY FLOOD INDEX

Normalized Difference Water Index (NDWI)

$$NDWI = (Green - NIR) / (Green + NIR)$$

Open water strongly absorbs NIR, so it tends to be dark in B8. Water can remain more reflective in the visible green band, creating a positive Green-NIR contrast. Positive NDWI often indicates open water, but a universal zero threshold should not be assumed across all land covers.

BACKGROUND CONCEPT

Normalized Burn Ratio (NBR)

$$NBR = (NIR - SWIR) / (NIR + SWIR)$$

Healthy vegetation typically has high NIR reflectance and lower SWIR reflectance because vegetation structure boosts NIR scattering while water in leaves absorbs strongly in SWIR. Fire damage reverses that contrast: NIR tends to decrease while SWIR increases, so NBR falls.

$dNBR = NBR_{pre} - NBR_{post}$ is widely used to map wildfire burn severity.

IMPORTANT INTERPRETATION RULE

Do not label flood pixels with wildfire burn-severity classes. The NBR background is useful because it explains why NIR-SWIR combinations are sensitive to vegetation and moisture disturbance. For Valencia, a moisture-oriented NIR-SWIR metric (for example NDMI) or a carefully defined change index would be more defensible for flood recovery than reusing dNBR severity thresholds.

WEBSITE-READY BLOCK

Suggested spectral explainer

Use a three-state interactive legend: water, healthy vegetation, dry/disturbed cover. Let visitors toggle Green, NIR, and SWIR responses, then reveal the NDWI and NBR equations only after the intuition is clear.

Conceptual reflectance logic

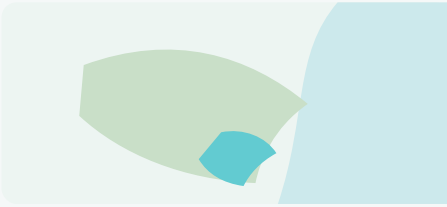


The strongest story is the spatial expansion and gradual recession of the water signature

BEFORE

Confined baseline water pattern

The source NDWI maps describe water signatures as largely restricted to La Albufera and other permanent water bodies, with much less apparent inundation across the surrounding agricultural flats.



IMMEDIATE POST-EVENT

Water signature expands inland

The post-event composite shows a much broader high-NDWI pattern extending across low-lying agricultural areas south of Valencia, consistent with widespread flooding and saturation after October 29.



RECOVERY

Recession toward baseline

The 2025 recovery maps show the water signature contracting. Some low-lying areas remain wetter than the pre-event pattern into summer, which may reflect persistent saturation or drainage constraints.



“The value of the time series is not just locating water once; it is showing how the spatial pattern departs from baseline and then returns.”

This interpretation follows the uploaded flood-mapping narrative. The diagrams above are original schematics and do not reproduce the source screenshots or mapped pixels.

EXTERNAL CONSISTENCY CHECK

CEMS independently documented extensive flood impacts in Valencia province, reporting more than 53,000 hectares affected as of October 31, 2024. This supports the broad regional severity of the event, but it is **not** a pixel-by-pixel validation of the source NDWI maps.

What the imagery supports - and what should remain a hypothesis

STRONGLY SUPPORTED

- Water is spectrally dark in NIR relative to many land surfaces.
- NDWI highlights a larger water / wet-surface footprint in the post-event period than in the pre-event windows.
- The strongest source-described changes occur in low-lying agricultural areas, channels, and the La Albufera system.
- The 2025 composites show a general movement back toward the pre-event pattern.

NEEDS VALIDATION OR QUALIFICATION

- High NDWI can indicate open water or very wet surfaces; one fixed threshold is not universally reliable.
- Long date-window composites can blur the short-lived maximum inundation footprint.
- Agricultural phenology, sediment/turbidity, built surfaces, shadows, cloud, and cloud shadow can affect index values.
- Persistent summer anomalies can suggest saturation or drainage problems, but imagery alone cannot attribute the cause to infrastructure damage.

HOW THE NBR BACKGROUND STRENGTHENS THE CASE STUDY

Use the spectral principle, not the wildfire label

The provided burn-severity material emphasizes that vegetation condition changes the balance between NIR and SWIR reflectance. That principle is valuable beyond fire: both vegetation water content and surface wetness influence these bands.

For a flood-recovery extension, the next step is to add a dedicated moisture / vegetation metric and compare it with NDWI. This can help distinguish **open water** from **wet soils** and **vegetation recovery**.

Question	Best signal	Interpretation	Caution
Where is open water?	NDWI	Green-NIR contrast	Threshold varies
Where is vegetation / canopy moisture changing?	NDMI or documented NIR-SWIR change	Moisture + vegetation response	Land-cover controls matter
How severe was wildfire damage?	NBR / dNBR	Burn severity	Not a flood class

WEBSITE-READY BLOCK

Make uncertainty visible. A small “what this map can / cannot tell you” panel will make the public-facing case study more credible and prevent index values from being interpreted as direct flood depth or damage.

Turn the descriptive mapping into a validated, quantitative flood-recovery analysis

01

Reconstruct peak inundation with Sentinel-1 SAR

Radar can observe through cloud and is well suited to event-day flood mapping. Compare SAR-derived water extent with CEMS rapid mapping and optical Sentinel-2 scenes.

02

Use shorter time steps and quantify recession

Replace broad multi-month composites with scene-level or 5-10 day summaries. Track flooded / wet area through time and estimate how quickly different zones return toward baseline.

03

Compare water indices and thresholds

Benchmark NDWI against alternatives such as MNDWI or AWEI. Calibrate thresholds by land cover rather than assuming a universal cutoff.

04

Add moisture and vegetation recovery

Use NDMI or a documented NIR-SWIR change metric to separate open-water recession from lingering soil / canopy moisture effects. Keep NBR/dNBR wildfire interpretation distinct.

05

Explain spatial controls

Overlay elevation, drainage networks, impervious cover, agricultural land use, and the La Albufera hydrologic system to test why some areas remain wet longer.

06

Move from hazard to exposure

Intersect validated flood extent with roads, buildings, cropland, and municipality boundaries to quantify who and what was exposed, then report uncertainty explicitly.

BEST NEXT DELIVERABLE

An interactive pre/post/recovery map slider with a synchronized area-over-time chart. It would preserve the intuitive visual story while adding quantitative evidence and uncertainty.

WEBSITE-READY BLOCK

Present these six items as a roadmap rather than unfinished work. Each can become a future interactive layer or a short "next question" card.

What this document uses - and what it deliberately does not reuse

Uploaded flood-mapping project PDF. Used for the Valencia study area's qualitative interpretation, the Sentinel-2 / Earth Engine workflow shown in the source, NDWI observations, and the five enumerated pre/post/recovery date windows.

Provided NBR and burn-severity background slides. Used only for concepts: healthy vegetation has high NIR and lower SWIR response; burned/dry disturbance lowers NBR; dNBR compares pre- and post-fire NBR; loss of vegetation can increase runoff and erosion risk after wildfire.

AEMET, "Situacion de lluvias intensas ... 28 de octubre - 4 de noviembre de 2024." Used to verify the meteorological context and Turis rainfall totals for October 29, 2024.

Copernicus Emergency Management Service, EMSR773. Used to verify activation, the regional flood context, and reported affected-area statistics for Valencia province.

Sentinel-2 User Handbook / product specifications. Used for MSI band center wavelengths and native spatial resolutions.

Google Earth Engine Data Catalog - COPERNICUS/S2_SR_HARMONIZED. Used to verify the surface-reflectance collection and common cloud-screening workflow.

U.S. Geological Survey burn-severity guidance. Used to verify $NBR = (NIR - SWIR)/(NIR + SWIR)$, $dNBR = pre - post$, and the correct wildfire-specific interpretation of burn severity.

U.S. Geological Survey literature citing McFeeters NDWI. Used to verify $NDWI = (Green - NIR)/(Green + NIR)$ and the general interpretation of positive values as open water.

VISUAL PROVENANCE

No source pictures are reproduced

All diagrams, maps, timelines, matrices, and icons in this PDF are newly drawn schematic graphics. They are explanatory, not substitutes for the original satellite imagery.

ACCURACY NOTE

The source narrative describes "four temporal windows" but lists five separate intervals. This document preserves the five actual intervals and flags the discrepancy rather than silently dropping one.

Recommended next production step: if this is converted into a website, replace schematic flood graphics with newly exported, properly attributed analysis layers and add a visible legend, time stamp, data source, and uncertainty note for every map.